the Journal of Strength and Conditioning Research™

Relative Cross-Education Training Effects of Male Youth Exceed Male Adults

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[1]Tunisian Research Laboratory "Sport Performance Optimisation" National Center of Medicine and Science in Sports, Tunis, Tunisia; [2]School of Human Kinetics and Recreation, Memorial University of Newfoundland, St. John's, Newfoundland and Labrador, Canada; [3]Department of Physical Education, Graduate Program in Health Sciences, Federal University of Sergipe, São Cristóvão, Brazil; [4]High Institute of Sport and Physical Education, Ksar-Said, Manouba University, Tunis, Tunisia; and [5]Sports Performance Research Institute New Zealand, AUT University, Auckland, New Zealand

Abstract

Ben Othman, A, Anvar, SH, Aragão-Santos, JC, Behm, DG, and Chaouachi, A. Relative cross-education training effects of male youth exceed male adults. *J Strength Cond Res* 38(5): 881–890, 2024—Cross-education has been studied extensively with adults, examining the training effects on contralateral homologous muscles. There is less information on the cross-education effects on contralateral heterologous muscles and scant information comparing these responses between adults and youth. The objective was to compare cross-education training effects in male youth and adults to contralateral homologous and heterologous muscles. Forty-two male children (10–13-years) and 42 adults (18–21-years) were tested before and following an 8-week unilateral, dominant or nondominant arm, chest press (CP) training program or control group (14 subjects each). Unilateral testing assessed dominant and nondominant limb strength with leg press and CP 1 repetition maximum (1RM), knee extensors, elbow extensors (EE), elbow flexors, and handgrip maximum voluntary isometric contraction (MVIC) strength and shot put distance and countermovement jump height. Upper-body tests demonstrated large magnitude increases, with children overall exceeding adults ($p = 0.05$—$p < 0.0001$, η^2: 0.51, 10.4 ± 11.1%). The dominant trained limb showed significantly higher training adaptations than the nondominant limb for the adults with CP 1RM ($p = 0.03$, η^2: 0.26, 6.7 ± 11.5%) and EE ($p = 0.008$, η^2: 0.27, 8.8 ± 10.3%) MVIC force. Unilateral CP training induced significantly greater training adaptations with the ipsilateral vs. contralateral limb ($p = 0.008$, η^2: 0.93, 27.8 ± 12.7%). In conclusion, children demonstrated greater training adaptations than adults, upper-body strength increased with no significant lower-body improvements, and ipsilateral training effects were greater than contralateral training in adults.

Key Words: children, boys, resistance training, strength, power

Introduction

Transferring the positive training effects of a trained limb (i.e., strength, endurance, skill, and acceleration) to the untrained limbs has been demonstrated repeatedly and is known as cross-education (26–28,49). This phenomenon has been demonstrated in different populations (17,23,24) with various physical and motor capacities (16). Unilateral training can induce both cross-education (contralateral, homologous muscles) and global (ipsilateral and contralateral, heterologous muscles) effects (4–6,13,14,44). However, there is a lack of consensus on the global effects between youth and adults using lower-limb or upper-limb training protocols.

For example, Evetovich et al. (18) showed a significant peak-torque improvement in both the trained and untrained quadriceps femoris following 12 weeks of leg extensions (6 sets of 10 leg extensions, 3 days per week). Adamson et al. (1) trained female adults' elbow flexors (EF) with their maximal intended velocity for 8 weeks and improved the rate of force development with a similar magnitude for both trained and untrained muscle groups. Boyes et al. (7) trained the subjects with high-frequency and low-frequency isometric handgrip training for 4 weeks (high frequency: 2 × 6 repetitions 10 times weekly; low frequency: 5 × 8 repetitions, 3 times weekly) and showed a significantly greater grip strength in both limbs posttraining. However, cross-education has also been reported with either small magnitude improvements (4) or no significant changes (5).

Cross-education and global training effects have been studied in youth (4–6,13,14,44). For example, following 8 weeks of unilateral leg press training thrice weekly, Ben Othman et al. (6) showed contralateral and ipsilateral homologous (leg press, knee extension–flexion isometric strength, and countermovement jump [CMJ]) and heterologous (EF and handgrip strength) positive adaptations in youth. In another study (5), youth subjects were trained with resistance training of their dominant upper (EF) and lower (knee extensors [KE]) limbs. However, unilateral upper-limb training led to significant improvements only with ipsilateral and contralateral upper-body testing. By contrast, unilateral leg press training improved only ipsilateral and contralateral lower-body measures (4). Chaouachi et al. (13) conducted a similar youth training study as this study using unilateral chest press (CP) and handgrip training with a similar testing protocol. They found that training to testing specificity with greater cross-education gains with the CP 1-repetition maximum (1RM) and handgrip maximum voluntary isometric contraction (MVIC) force specifically when training with CP and handgrip, respectively. However, only the dominant limb was trained in

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Journal of Strength and Conditioning Research 38(5)/881–890

that study, and thus, differences in unilateral training of the dominant vs. nondominant limb need to be investigated in youth and adult populations as well.

Neural mechanisms have been suggested to be involved in cross-education (27,49). Bilateral coactivation through corticospinal pathways (11), impulse diffusion between cerebral hemispheres (12), coordination learning or postural stabilization (9), and modulation of afferents (29) have all been suggested as possible mechanisms. Because children are less experienced with various activities, neuromuscular learning (i.e., motor control) could provide a higher contribution to the contralateral strength gains than with adults (2). Cross-education strength training gains have also been recorded following 6 weeks of 1–2 hours daily plantar flexors stretching and attributed to central neural adaptations because of stimulation of the stretched muscles (47,48). Hence, high-intensity resistance training is not the only pathway to improved cross-education training effects. However, according to the scoping review by Zhou et al. (50), mechanisms underlying the contralateral effects of unilateral peripheral stimulation are unclear. However, despite these mechanisms, the cross-education effects are not consistent in different populations and training protocols.

Thus, with few studies investigating the global effects on adults, it is unknown whether the global or cross-education effects are similar between youth and adults. Also, the literature has little consensus about the possible cross-education or global effects on upper-limb training protocols and possible differences when training the dominant vs. nondominant arm. Hence, it is worthwhile to investigate whether unilateral training similarly induces global effects in youth and adults to understand the optimal training approach in both populations. Theoretically, differences in global or cross-education effects can imply age-related differences in overall strength training neural adaptations. In practical terms, age-related cross-education differences could influence the extent and diversity of training exercises needed to achieve full-body training adaptations. Therefore, this study aimed to determine whether there were cross-education differences between youth and adults in response to an 8-week, unilateral, dominant-arm, CP training program. We hypothesized that both groups would show cross-education effects. However, it was hypothesized that youth could show more pronounced cross-education effects.

Methods

Experimental Approach to the Problem

Two weeks before the commencement of the study, all subjects (children and adults) participated in 5 orientation sessions to become familiar with the general environment, form, and technique of training exercise and tests (force and power techniques), equipment, and the experimental procedures to minimize the learning effect. Each subject's height, sitting height, and body mass were collected using a wall-mounted stadiometer and electronic scale (LifeSource Model UC-321P, made by A&D Company, Tokyo, Japan), respectively. Afterward, subjects' performances were tested before and after the 8-week unilateral CP training period. The testing protocol and primary outcome included the assessment of lower-limb, unilateral, dominant and nondominant limb strength (force output) with horizontal leg press 1RM, KE MVIC, upper-body unilateral CP 1RM, MVIC strength of elbow extensors (EE) and EF, handgrip MVIC strength, and proxies of muscle power (shot put distance and CMJ). Following the initial baseline testing session, subjects from each age group were randomly divided into 2 unilateral CP resistance training groups (dominant-limb CP training, $n = 14$; nondominant-limb CP training, $n = 14$) and a control group ($n = 14$) without a training program. Footedness and handedness were assessed by Waterloo Handedness and Footedness questionnaires, respectively, to determine the dominant upper and lower limb. Using a controlled randomization method, groups were matched for age, maturation status, and physical characteristics. The 8-week (3 sessions per week) training program was periodized, and the volume of work during training was equal between the experimental groups, as previously described (4–6,13).

Subjects

Based on prior related cross-education research (5,6,13,14), an "a priori" statistical power analysis was calculated based on force data, determining that 62 subjects would be needed to achieve moderate magnitude changes, considering an effect size f of 0.5, alpha of 0.05, and a power of 0.8 for a fixed-effects analysis of variance (ANOVA) for main effects and interactions (noncentrality parameter $\lambda = 15.5$, critical $F = 2.2687175$, denominator df = 55). Hence, 42 male children between 10 and 13 years, from the same primary public school (Bou-Arada city), and 42 healthy male adults aged between 18 and 21 years, recruited from the same secondary school (Farat Hachad, Bouarada City, Tunisia), volunteered to participate in this study (Table 1). The 42 subjects in each group were divided into groups of 14 (dominant and nondominant CP training and control groups). All subjects were from similar socioeconomic status, and each age group had similar daily school schedules. All of them participated in their normal physical education lessons, and none were involved in any after-school activities or any formalized strength and conditioning training programs. A maturity status assessment was conducted using the noninvasive technique proposed by Mirwald et al. (42). All

Table 1
Subject characteristics.*

Training groups	Age (y)	Body mass (kg)	Height (cm)	BMI (kg·m^{-1})	PHV (y)
Prepubertal male children					
Dom chest press ($n = 14$)	12.2 ± 0.8	40.5 ± 6.6	147.1 ± 4.5	18.6 ± 2.2	−2.1 ± 0.4
Non-Dom chest press ($n = 14$)	11.6 ± 0.4	39.9 ± 5.5	146.1 ± 4.3	18.7 ± 2.5	−2.3 ± 0.4
Control ($n = 14$)	11.7 ± 0.6	41.5 ± 7	147.3 ± 4.2	19.1 ± 3.1	−2.3 ± 0.4
Young male adults					
Dom chest press ($n = 14$)	18.8 ± 0.6	178.2 ± 6.7	77.9 ± 9	24.5 ± 2.5	N/A
Non-Dom chest press ($n = 14$)	19.3 ± 0.8	176.3 ± 6.4	77.8 ± 8.4	25.0 ± 2.2	N/A
Control ($n = 14$)	19.1 ± 0.6	178.7 ± 5.8	78.1 ± 6	24.5 ± 2.3	N/A

*Dom = dominant; Non-Dom = nondominant; BMI = body mass index; PHV = peak height velocity; N/A = not applicable.

child subjects were classified at the prepeak height velocity stage of physical maturation (Table 1).

Parent or guardian (for children) and subject informed consent was obtained after thorough explanation of the objectives and scope of this project, the procedures, risks, and benefits of the study, before any data collection. The study was conducted according to the latest version of the Declaration of Helsinki, and the protocol was approved by the Manouba University, local Ethics Committee of the National Centre of Medicine and Science of Sports of Tunis (CNMSS-LR09SEP01) before the commencement of the assessments. Subjects and their parent or guardian were also informed that participation was voluntary and that they could withdraw from the study at any time. None of the subjects had any history of musculoskeletal, neurological, or orthopedic disorders that might impair their ability to execute resistance exercise training or to perform strength and power tests.

Procedures

Training Programs. Similar to the previous series of youth cross-education or global training studies (4–6,13) from this laboratory, the subjects trained for 8 weeks, completing 3 sessions per week with at least 48–72 hours of rest between the sessions (totaling 24 sessions). Three sets of 8, 9, 10, and 6 repetitions were completed in weeks 1–4, respectively, followed by 4 sets of 8, 9, and 10 repetitions in weeks 5–7, respectively, with a reduction to 3 sets of 6 repetitions in the final week. Two-minute recovery was allowed between the sets for both training programs. Two to 5 days after the last training session, posttraining was performed using the same time line and procedures as during the pretest (4–6,13). Progressive overload of the dominant-arm CP group was implemented by increasing the load by 5–10% whenever a subject could exceed the prescribed 6–10 repetitions with proper form. All trained subjects attended at least 22 of the 24 training sessions (missed no more than 2 sessions), and none of the subjects reported any training or test-related injury. The control group was limited to their regular daily activity (no structured or systematic training or activity).

Before each training session, the training groups performed a specific warm-up consisting of submaximal ergometer cycling for 5 minutes before dynamic stretching. The exercise used for training was a unilateral CP in a seated position with either the dominant or nondominant arm with a range of motion from 90 to 10°. The untrained contralateral arm was placed across the chest during the resistance exercise repetitions to ensure that subjects did not provide a strength training stimulus to the untrained arms or hands by stabilizing their upper body. The heels of the feet were placed on a 20-cm box to avoid generating ground reaction forces. The sets were performed to momentary failure, defined as the inability to complete a full repetition despite the greatest effort in attempting to do so. Subjects in the training groups received skill-specific feedback on the quality of each movement. The instructors recorded the training data on workout logs daily and made appropriate adjustments in training resistance and repetitions. Special attention was paid to the instructions to keep the contralateral arm and both legs completely immobile and as relaxed as possible during the training.

Lower-Body Maximal Strength and Power Tests

Unilateral, Leg Press, Maximal Dynamic Strength (1RM). Unilateral leg strength was assessed on the horizontal leg press with a 1RM test. Before attempting a 1RM, subjects performed 3 submaximal sets of 1–6 repetitions with a light-to-moderate load, then 3 sets of a heavier load. Finally, subjects performed a series of single repetitions with increasing loads. If the weight was lifted with the proper form, it was increased by approximately 1–2 kg, and the subject attempted another repetition. The increments in weight were dependent on the effort required for the lift. One repetition maximum was defined as the greatest load lifted through a full ROM before 2 failed attempts at a given load. The exercise execution technique was standardized and continuously monitored in an attempt to assure the quality of the data. The subjects were strapped into the apparatus with a seat belt with the nonexercised leg positioned off the leg press apparatus (foot on the floor) in a relaxed state. Subjects folded their arms across their chest during the procedure. Throughout all testing procedures, an instructor-to-subject ratio of 1:1 was maintained, and uniform verbal encouragement was offered to all subjects.

Unilateral Knee Extensor Maximum Voluntary Isometric Contraction. Maximal voluntary isometric KE strength was measured in both the dominant and nondominant limbs using a calibrated handheld, load cell dynamometer (muscle controller, Kinvent Biomechanique, Montpelier, France). Specifics of the test position, stabilization, and dynamometer placement used in this study were chosen according to the instrument manual instructions, as previously described (4–6,13). The handheld dynamometer (HHD) was placed perpendicular to the anterior aspect of the tibia, just proximal of the medial malleolus for quadriceps testing. Subjects were seated on the chair of the leg extension machine (Life Fitness Pro Horizontal Leg Press), positioned so that both feet were off the ground, with hips and knees both flexed at 90°. The lever arm of the leg extension machine was fixed at 100°. The dynamometer was fixed and stabilized by the examiner between the lever arms of the machine and the specific testing placement position on the tested limb segment. The arm of the leg extension machine was fixed with a maximal load to ensure that subjects performed an isometric contraction. Subjects were instructed to exert maximal force against the dynamometer for a period of 3 seconds. Three consecutive trials separated by approximately 1 minute for both legs and the highest values were recorded for analysis. The same researcher performed all handheld dynamometry measures. High handheld dynamometry reliability measurements in a similar pediatric population in our laboratory have been reported elsewhere (4–6,13).

Unilateral Countermovement Jump. The unilateral (single-leg) CMJ test was performed using an Ergo Jump system (Ergojump: Globus Italia, Codogne, Italy) according to the procedure described previously (4–6,13). Subjects started from an upright akimbo position. Subjects self-selected the amplitude of the knee flexion of the CMJ to avoid changes in the coordination pattern. The nonjumping leg was held in a slightly flexed relaxed position during the unilateral jump. Three trials were performed for each leg with approximately 1 minute of recovery between the trials, and the highest jump was used for analysis. High reliability of this test (intraclass correlation coefficients [ICC] = 0.95) in a similar pediatric population in our laboratory has been published previously (4–6,13).

Unilateral Isometric Strength (Knee Extensors, Elbow Flexors, and Elbow Extensors Maximum Voluntary Isometric Contraction). Maximal voluntary isometric EF and EE strength of both arms were measured using a calibrated HHD (muscle controller, Kinvent Biomechanique, Montpelier, France).

Specifics of the test position, stabilization, and dynamometer placement used in this study were chosen according to the instrument manual instructions, as previously described (13). For elbow flexion, the HHD was placed between the flexor aspect of the wrist and the lever of the elbow flexion machine to ensure an isometric contraction. For elbow extension, the HHD was placed against the wrist (distal part of the rear face of forearm) by the researcher. The elbow was flexed at 90°. Next, subjects were instructed to flex or extend their elbow with maximal force for 3 seconds. This procedure was repeated 3 times for both the right and left hands with an approximate 1-minute rest period, and the highest value was recorded for analysis. High reliability (0.84–0.92) of this test in a similar pediatric population in our laboratory has been reported elsewhere (4–6,13).

Unilateral Handgrip Maximum Voluntary Isometric Contraction. Maximum voluntary isometric contraction handgrip strength (in kilograms) was measured using a calibrated hand dynamometer (Takei, Tokyo, Japan), as previously described (4–6,13). Subjects stood with the arm adducted at approximately 45°. The dynamometer was held freely without support and did not touch the subject's trunk, with constant extension of the elbow. The grip span of the dynamometer was adjusted to each subject's hand size so that the proximal interphalangeal joints of the 4 fingers rested on one side of the handgrip and that of the thumb rested on the other side. Three trials separated by an approximate 1-minute rest interval for each hand were performed, and the maximum score for each hand was recorded. Excellent MVIC handgrip strength reliability measurements in children in our laboratory have been reported elsewhere (ICC = 0.78–0.91) (4–6,13).

Unilateral, Seated, Single-Arm, Shot Put Test. Subjects were asked to hold a medicine ball anterior to the shoulder with the elbow adjacent to the torso and keeping their elbow tucked against their torso and as far back against the wall as possible. The nontested arm was positioned in the lap, and the subject attempted to "put" a 3-kg medicine ball the greatest distance. The initial location of medicine ball ground contact was measured relative to the starting position and reflects the test limb's underlying functional strength and power (4–6). Test trials were repeated if subjects crossed the torso midline with their testing arm, moved the torso away from the wall, bent their knees, or preloaded before completing the test. The best of 3 trial for each arm was recorded.

Statistical Analyses

Data were analyzed using the software Statistical Package of Social Sciences version 28 (SPSS Inc., Chicago, IL). The assumption of normality was tested and confirmed with the Shapiro-Wilk test, and the sphericity was assumed for all the dependent variables with the Mauchly's test. In the case that the sphericity was violated, the corrected value for nonsphericity with Greenhouse-Geisser epsilon was reported. A 1-way repeated-measures ANOVA was used to compare time effects (pretest to posttest). Then, relative (pretest to posttest % changes) training-related changes were calculated and implemented into a 3-way ANOVA to compare 3 training groups (unilateral CP training with dominant arm, nondominant arm, and controls); × 2 ages (children and adults) × 2 tested limbs (dominant and nondominant) to analyze possible relative (%) training adaptations with chest and leg press 1RM, EE, EF, KE, and handgrip MVIC forces, shot put distance, and CMJ height. Bonferroni's adjusted post hoc tests were used to locate the specific main effect differences, whereas post hoc t-tests were used to identify specific differences with significant interactions. η^2 effect size guidelines suggest that η^2 values of ≥0.01, ≥0.06, and ≥0.14 represent small, medium, and large magnitude effects, respectively (34). Reliability was assessed with Cronbach's alpha ICC with coefficient of variations reported.

Results

There were no significant improvements in any of the lower-body measures (Table 2).

Reliability

Test to re-test values demonstrated excellent reliability ranging from 0.934 to 0.993 (Table 3).

Adult vs. Child Differences

All upper-body tests demonstrated significant (except for the handgrip MVIC: $p = 0.08$), large magnitude, training increases for the children, with child exceeding adult means by 10.4% (Table 4 and Figure 1). There were no significant main effects for age differences for any of the lower-body tests (i.e., leg press 1RM, KE MVIC, CMJ).

Trained Limb Differences

All upper-body tests demonstrated significant, large magnitude, training increases for both trained limbs vs. controls with dominant and nondominant trained limbs generally exceeding control means by 23.1 and 18.5%, respectively (Table 5 and Figure 2). There were no significant main effects for trained limb (dominant vs. nondominant vs. control) differences for any of the lower-body tests (i.e., leg press 1RM, KE MVIC, CMJ).

Table 2
Lower-body statistics.*

	Leg press 1RM	Knee extension MVIC	Countermovement jump
Main effect for age	$F_{(1,13)} = 0.42, p = 0.53$	$F_{(1,13)} = 0.14, p = 0.72$	$F_{(1,13)} = 0.004, p = 0.95$
Main effect for training	$F_{(2,26)} = 1.22, p = 0.31$	$F_{(2,26)} = 0.52, p = 0.59$	$F_{(2,26)} = 1.98, p = 0.16$
Main effect for limb dominance	$F_{(1,13)} = 0.048, p = 0.83$	$F_{(1,13)} = 0.55, p = 0.47$	$F_{(1,13)} = 1.18, p = 0.29$
Age × training interaction	$F_{(2,26)} = 0.08, p = 0.92$	$F_{(2,26)} = 0.45, p = 0.65$	$F_{(2,26)} = 0.54, p = 0.59$
Training × testing interaction	$F_{(2,26)} = 0.18, p = 0.84$	$F_{(2,26)} = 1.08, p = 0.35$	$F_{(2,26)} = 0.72, p = 0.49$

*1RM = 1 repetition maximum, MVIC = maximum voluntary isometric contraction.

Table 3
Intraclass correlation coefficient (ICC) reliability and coefficients of variation.*

Measure	Pretest	Retest	ICC value	Coefficient of variation (%) Test/retest
CP 1RM Dom (kg)	38.5 ± 6.97	38.5 ± 7.51	0.982	6.97/7.51
CP 1RM non-Dom (kg)	38.7 ± 6.85	38.1 ± 6.66	0.979	6.85/6.66
EF MVIC Dom (kg)	32.9 ± 5.28	33.2 ± 5.60	0.934	5.25/5.60
EF MVIC non-Dom (kg)	33.5 ± 4.86	33.9 ± 5.56	0.967	4.86/5.56
EE MVIC Dom (kg)	19.7 ± 4.94	19.5 ± 4.76	0.978	4.94/4.76
EE MVIC non-Dom (kg)	20.8 ± 4.90	20.7 ± 4.79	0.987	4.90/4.79
HG Dom (kg)	42.6 ± 8.62	42.6 ± 8.84	0.989	8.62/8.44
HG non-Dom (kg)	40.2 ± 8.10	39.4 ± 7.81	0.970	8.10/7.81
Shot put Dom (cm)	519.9 ± 63.71	518.3 ± 61.38	0.992	12.25/11.84
Shot put non-Dom (cm)	487.2 ± 47.65	486.3 ± 48.93	0.987	9.78/10.06
KE MVIC DOM (kg)	45.3 ± 7.25	44.9 ± 7.11	0.991	7.25/7.11
KE MVIC non-Dom (kg)	44.5 ± 7.27	44.0 ± 6.51	0.984	7.27/6.51
LP 1RM DOM (kg)	89.6 ± 13.91	89.8 ± 14.30	0.990	13.91/14.3
LP 1RM non-Dom (kg)	89.9 ± 14.37	89.3 ± 13.89	0.993	15.99/15.55
CMJ Dom (cm)	15.7 ± 2.78	15.6 ± 2.71	0.985	17.74/17.44
CMJ non-Dom (cm)	15.9 ± 3.35	15.9 ± 3.20	0.987	21.13/20.12

*CP = chest press; 1RM = 1-repetition maximum; Dom =dominant limb; Non-Dom = nondominant limb; EF = elbow flexors; EE =elbow extensors; HG = handgrip; KE = knee extensors; MVIC = maximum voluntary isometric contraction; LP = leg press; CMJ = countermovement jump.

Tested Limb Differences

Only shot put testing identified a significant main effect for the tested limb with a greater training response for the nondominant (12.3%) over the dominant (9.4%) tested limb.

Age × Training Interactions

The dominant trained limb showed higher training adaptations than the nondominant limb only for the adults with CP 1RM and EE MVIC force (Table 6). Chest press 1RM and EE MVIC dominant arms exceeded nondominant arms by 6.7% ($p = 0.04$) and 8.8% ($p = 0.003$), respectively. Both trained limbs exceeded the control group for both adult and children CP 1RM and EE MVIC (all differences $p < 0.0001$; Table 6).

Trained Limb × Testing Limb Interactions

Unilateral dominant CP training induced significant training adaptations with the ipsilateral limb (DOM-DOM) over the contralateral (DOM-nonDOM) limb for CP 1RM, EE MVIC, EF MVIC, and shot put (Table 7 and Figure 3). Similarly, there was significant training adaptations with unilateral nondominant CP training when testing the ipsilateral limb (nonDOM-nonDOM) vs. the contralateral limb (non-DOM-DOM) for CP 1RM, EE MVIC, EF MVIC, handgrip MVIC, and shot put (Table 7 and Figure 3). The extent of ipsilateral training adaptations was significant when training the dominant limb (DOM-DOM) vs. the nondominant limb (nonDOM-nonDOM) with the EE MVIC (Table 7 and Figure 3).

Discussion

The major findings of this study were that following unilateral CP training: (a) children demonstrated relatively greater training adaptations vs. adults, (b) all upper-body strength measures increased, although there were no significant improvements with lower-body testing, (c) testing limb specificity was evident with ipsilateral training effects (dominant limb training and testing [DOM-DOM]; nondominant training and testing [nonDOM-non-DOM]) generally (data combined for adults and children) greater than contralateral training effects (dominant limb training with nondominant limb testing: DOM-nonDOM and nondominant limb training with dominant limb testing: nonDOM-DOM), and (d) however, children did not exhibit significant differences between ipsilateral and contralateral training effects.

The finding that children's upper limb adaptations were greater (10.4%) with unilateral CP training compared with adults in this study was similar to the results of Chaouachi et al. (13) who showed that there were greater contralateral, nondominant arm, CP training improvements in youth compared with adults (35.8% youth vs. 21.5% adults). By contrast, 2 studies reported no

Table 4
Main effects for age with percentage increases illustrated from pretest to posttest and effect size magnitudes (η^2).*†

Measures	Significance	η^2	Children, %	Adults, %
Unilateral CP	$F_{(1,13)} = 54.69$, $p < 0.0001$	0.808	38.5 ± 18.4	20.1 ± 10.4
EE MVIC	$F_{(1,13)} = 72.82$, $p < 0.0001$	0.829	38.2 ± 12.6	19.7 ± 11.6
EF MVIC	$F_{(1,13)} = 4.17$, $p = 0.05$	0.218	18.3 ± 8.5	11.4 ± 8.1
Handgrip MVIC	$F_{(1,13)} = 3.56$, $p = 0.08$	0.215	9.2 ± 7.8	6.0 ± 6.2
Shot put	$F_{(1,13)} = 12.51$, $p = 0.004$	0.491	13.6 ± 14.8	8.2 ± 6.0
Means		0.512	23.5	13.1

*CP = chest press; EE = elbow extension; MVIC = maximal voluntary isometric force; EF = elbow flexors.
†Means ± *SD*s are presented for percentage changes.

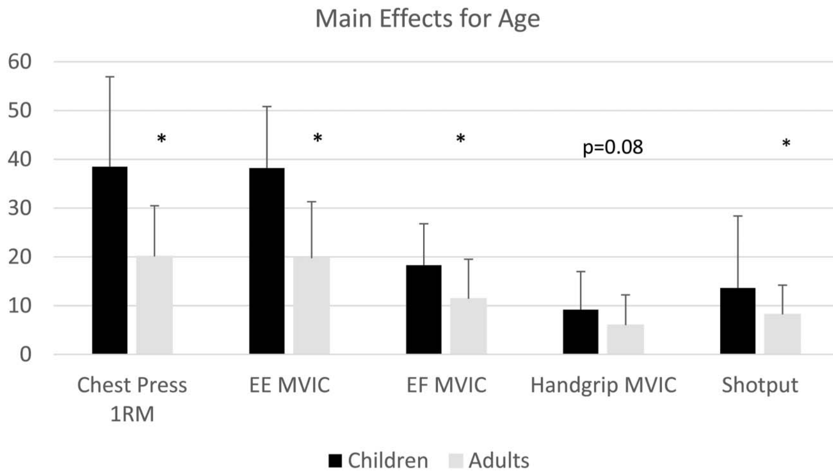

Figure 1. Main effects for age with percentage increases illustrated from pretest to posttest and effect size magnitudes (η^2). Means ± *SDs* are presented for percentage changes. Asterisks (*) highlight significant differences ($p < 0.05$).

significant cross-education, isometric strength gain differences between young (28.2 ± 3.1) and elderly women (73.1 ± 5.3 years) (17) nor young (20.8 ± 0.1 years) and older women (58.1 ± 0.14 years) (3), albeit the young women in these studies were young adults not youth.

Training with greater complexity can induce greater motor learning or coordination adaptations with the capability of transfer to uninvolved muscles (38). Sufficient training complexity with a large muscle group such as with CP training (13,14) especially in naive subjects with little to no experience resistance training may help explain the greater magnitude and extent of cross-education effects in this study. Similarly, in another study training large lower-limb muscles such as with leg press training induced extensive homologous (cross education) and upper-body (EFs and handgrip strength) muscle strength gains that were approximately 10% of the gains observed with the trained muscles (6). They (6) suggested that youths' central nervous system may be more adaptable than that of adults, permitting a more generalized, global training response.

The improvement of the upper-body measures and not the legs in this study agrees with the results of Ben Othman et al. (5,6). However, Chaouachi et al. (13) reported significantly greater lower-body gains in adults than in youth following dominant arm, unilateral CP training. There are also contradictions when examining review articles. Meta-analysis of Manca et al. (39) revealed significantly greater cross-education gains with adult (23.9 ± 3.3 years) lower limbs (16.4%) vs. upper limbs (9.4%). However, another meta-analysis reported similar cross-education training effects with both the upper and lower limbs in young (<50 years) and older (>50 years) adults (25). Considering the greater prevalence of the central nervous system (CNS) locomotor pattern generation in the legs than in the arms (5), the global adaptations to unilateral upper-body or lower-body training in youth remain to be clarified with these conflicting results.

This study results agree with those of Chaouachi et al. (13), showing significantly greater training gains in CP 1RM than handgrip following unilateral CP training of the dominant arm. They showed that adults and youth improved their dominant arm's CP 1RM by 50.3 and 81.1% following the dominant arm's CP training. Many studies demonstrated greater cross-education effects with the dominant limb training (21,31,45). By contrast, Razian et al. (44) found greater cross-education training gains with the nondominant leg following 12 weeks of unilateral training of middle-aged women. However, Coombs (15) reported symmetrical cross-education effects in adults after dominant or nondominant training. Coombs et al. suggested the task and training model differences between their study and others as a contributing factor (15).

In agreement with this study, Ben Othman (6) tested the dominant leg following the nondominant leg's training and vice

Table 5

Main effects for trained limb (dominant vs. nondominant, vs. control).*†

Measures	Significance	η^2	Dominant, %	Nondominant, %	Control, %
Unilateral CP	$F_{(2,26)} = 61.18$, $p < 0.0001$	0.825	45.4 ± 11.5	37.8 ± 13.1	4.5 ± 10.1‡
EE MVIC	$F_{(2,26)} = 80.45$, $p < 0.0001$	0.843	45.9 ± 11.1	37.1 ± 10.4§	3.9 ± 8.8‡
EF MVIC	$F_{(2,26)} = 8.56$, $p = 0.001$	0.364	21.2 ± 11.3	16.7 ± 7.5	6.6 ± 15.5‡
Handgrip MVIC	$F_{(2,26)} = 7.07$, $p = 0.004$	0.325	9.9 ± 7.2	9.6 ± 6.8	3.2 ± 7.5‡
Shot put	$F_{(2,26)} = 19.91$, $p < 0.0001$	0.605	15.2 ± 6.2	13.5 ± 10.2	4.0 ± 7.9‡
Means		0.592	27.5	22.94	4.4

*CP = chest press; EE = elbow extension; EF = elbow flexors; MVIC = maximal voluntary isometric force.
†Means ± *SDs* are presented for percentage changes.
‡Increase in the control group was significantly less than both the dominant and nondominant limb training improvements.
§Nondominant trained limb experienced significantly lower relative (%) positive training adaptations than the dominant limb.

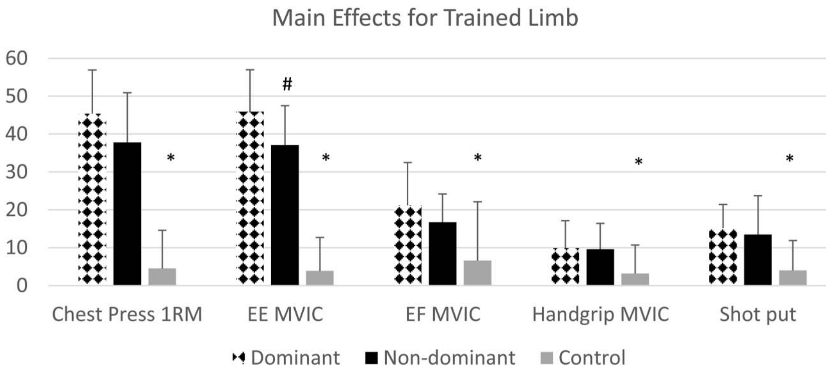

Figure 2. Main effects for trained limb (dominant vs. nondominant, vs. control). Asterisks (*) illustrate that the increase in the control group was significantly less than both the dominant and nondominant limb training improvements. The hashtag (number #) symbol highlights that the nondominant trained limb experienced significantly lower relative (%) positive training adaptations than the dominant limb. Means ± *SD*s are presented for percentage changes.

versa, showing that children did not demonstrate greater gains with the dominant limb. They reported that both dominant and nondominant leg press training significantly improved lower-body strength factors ipsilaterally and contralaterally. Not observing a directionality in transferring the cross-education effects in children suggests that the children's CNS may be more adaptable than that of adults (2). However, the specific location of this process in the motor-controlling system cannot be confirmed in this study.

In other studies, conducted with adults, the lack of increases in the electromyography activity with cross-education suggests that cross-education may be more attributed to the increased activation of motor cortex (30). Cross-education has been suggested to be related to cortical pathways' plasticity involved in motor planning output and plasticity in the motor command of the motor cortex, increasing agonist activation (22,28) and decreasing cocontractions (10).

It is ubiquitously postulated that because anabolic endocrine responses to signal muscle hypertrophy are substantially lower in prepubertal boys than in adult men, neural adaptations are the primary mechanism underlying strength adaptations (2,19,20). Neuromuscular factors like increased corticospinal excitability (32), decreased corticospinal inhibition (i.e., short-interval cortical inhibition) (35), decreased interhemispheric inhibition, enhanced motor unit recruitment, and rate coding have been postulated to contribute to improved contralateral limb power and strength (22,36,40,49). However, this could not be confirmed with meta-analyses showing a poor correlation between the magnitude of corticospinal excitability with cross-education changes (39,40). Lack of H-reflex changes of the contralateral untrained soleus of adults after 5 weeks of strength training suggested that the cross-education effect may be due more to supraspinal mechanisms (33). Because children are less experienced, their degree of neuromuscular learning (i.e., motor control) would provide a higher contribution to the contralateral strength gains than that with adults (2). A level of unintended movements following a unilateral activity has also been reported in homologous muscles on the opposite side of the body (41). This unintentional movement activity, the so-called mirror movement, is shown to decrease as a child matures (41). More uncrossed, ipsilateral, corticospinal projection in children controlling the voluntary activated hand has been suggested (43). Mirror movements may also be produced by bilateral cortical activity, possibly because of a lack of callosal inhibition (37). Youth neural development is differentiated from adults by greater increases in myelination, synaptic organization, and pruning (especially with higher cognitive functions with adolescents) (46). Hence, further investigations are still needed to pinpoint possible neural adaptations.

Table 6

Age × training interactions.*†

Measures	Significance	η^2	Dominant arm, %	Nondominant arm, %	Control, %
Chest press 1RM	$F_{(2,26)} = 4.71$, $p = 0.037$	0.266			
Adult			32.8 ± 15.9	26.1 ± 7.1§	1.3 ± 8.3‡
Children			58.1 ± 22.6‖	49.6 ± 8.2‖	7.7 ± 10.4‡
EE MVIC	$F_{(2,26)} = 5.66$, $p = 0.008$	0.274			
Adults			33.7 ± 11.3	24.9 ± 9.3§	0.5 ± 5.2‡
Children			58.2 ± 12.3‖	49.2 ± 11.9‖	7.3 ± 9.9‡
Means		0.270	45.7	37.5	4.2

*EE =elbow extensors; MVIC = maximal voluntary isometric contraction; 1RM = 1-repetition maximum.
†Means ± *SD*s are presented for percentage changes.
‡Increase in the control group was significantly less than both the dominant and nondominant limb training improvements.
§Nondominant trained limb experienced significantly lower relative (%) positive training adaptations than the dominant limb.
‖Children had significantly higher relative increases than the adults.

Table 7

Trained Limb × Testing Limb interactions.*†

Measures	Significance	η^2	(IPSI) DOM DOM	(CONTR) DOM Non-DOM	(CONTR)Non-DOM DOM	(IPSI) Non-DOM Non-DOM	Control DOM	Control Non-DOM
Chest press 1RM	$F_{(2,26)} = 188.3$, $p = 0.008$	0.93	59.3§ ±15.9%	31.5 ±9.5%	23.8 ±7.1%	51.8 ±9.3%‡	4.6 ±9.3%‖	4.5 ±8.3%‖
EE MVIC	$F_{(2,26)} = 218.7$, $p < 0.0001$	0.93	37.1§ ±11.3%¶	22.2 ±15.2%	19.0 ±10.7%	30.0 ±9.1%‡	4.4 ±9.3%‖	4.2 ±5.2%‖
EF MVIC	$F_{(2,26)} = 21.6$, $p < 0.0001$	0.59	18.1 ±10.8%§	12.5 ±8.3%	7.6 ±4.4%#	19.7 ±8.7%‡	5.8 ±7.5%‖	5.8 ±4.9%‖
Handgrip MVIC	$F_{(2,26)} = 5.6$, $p = 0.009$	0.30	10.5 ±4.1%	9.2 ±5.2%	7.3 ±9.3%	11.8 ±6.2%‡	2.1 ±7.0%‖	4.2 ±3.7%‖
Shot put	$F_{(2,26)} = 5.4$, $p = 0.037$	0.29	18.3 ±8.9%§	12.1 ±6.5%	6.9 ±4.0%	20.0 ±4.7%‡	3.1 ±4.6%‖	4.8 ±10.9%‖
Means		0.61	28.6%	17.5%	12.9%	26.6%	4.0%	4.7%

*EE =elbow extensors; MVIC = maximal voluntary isometric contraction; EF = elbow flexors; 1RM = 1 repetition maximum; DOM-DOM = dominant limb training with testing of the dominant limb (ipsilateral effects); DOM-non-DOM = dominant limb training with testing of the nondominant limb (contralateral effects); non-DOM-DOM = nondominant limb training with testing of the dominant limb (contralateral effects), non-DOM-non-DOM: nondominant limb training with testing of the nondominant limb (ipsilateral effects).

†Means ± *SD*s are presented for percentage changes.

‡Significantly greater ipsilateral (non-DOM-non-DOM) vs. contralateral (non-DOM-DOM) between the tested limbs with unilateral nondominant CP training.

§Significantly greater ipsilateral (DOM-DOM) vs. contralateral (DOM-non-DOM) training adaptations between the tested limbs with unilateral dominant CP training.

‖Significantly lower relative increases with control vs. trained groups.

¶Significantly larger relative increase with ipsilateral effects of dominant (DOM-DOM) vs. nondominant (non-DOM-non-DOM) limbs.

#Significantly larger relative increase with contralateral effects of dominant (DOM-non-DOM) vs. nondominant (non-DOM-DOM) limbs.

Predominance of adults' dominant-limb training effect observed in the studies (21,31) may be related to the stronger neural networks because of decades of task preference in the dominant limb (6). However, limb preference does not seem well established with youth (8). This results in a less firmly established neural network in children (6). Thus, cross-education may have more limb equity in youth.

It is necessary to consider some limitations of this study, such as the training protocol based on a specific exercise (i.e., CP). Although it was not the aim of this study to examine sex differences, it is a common problem in science that female subjects are underrepresented, and thus, the absence of female subjects make the generalizability of these findings to other exercises and female subjects more difficult. Conversely, a higher number of male subjects improves data homogeneity.

In conclusion, following unilateral CP training, children demonstrated relatively greater training adaptations than adults. Upper-body strength measures significantly increased contrasting with no significant improvements with lower-body testing. In adults, limb specificity was evident with ipsilateral training effects greater than contralateral training effects. But this was not observed with the children. Thus, cross-education may be a stronger and more global phenomenon with children than with adults.

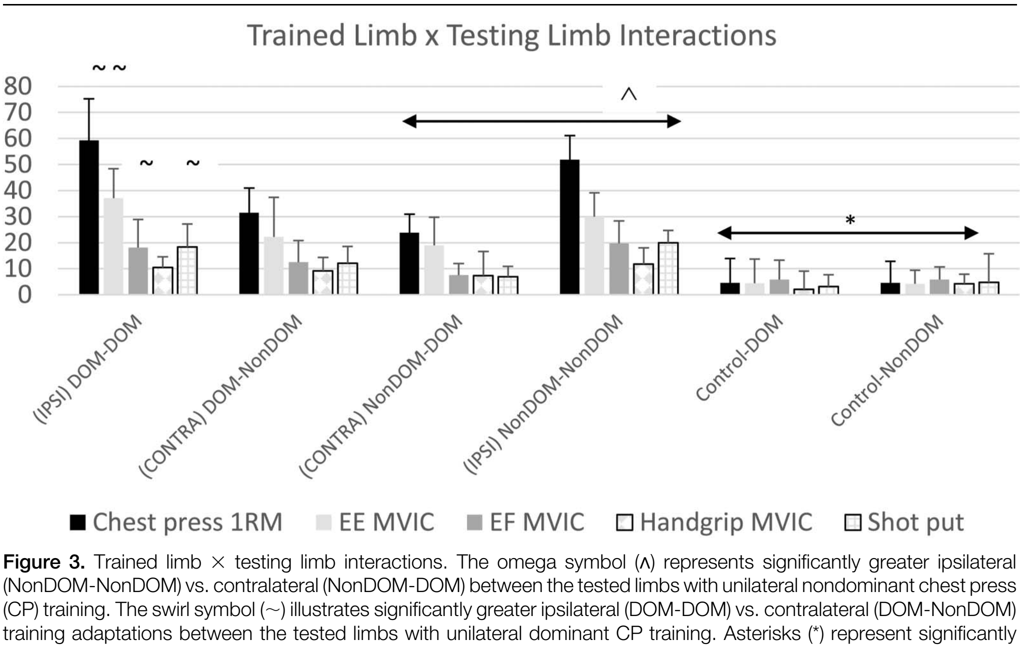

Figure 3. Trained limb × testing limb interactions. The omega symbol (∧) represents significantly greater ipsilateral (NonDOM-NonDOM) vs. contralateral (NonDOM-DOM) between the tested limbs with unilateral nondominant chest press (CP) training. The swirl symbol (~) illustrates significantly greater ipsilateral (DOM-DOM) vs. contralateral (DOM-NonDOM) training adaptations between the tested limbs with unilateral dominant CP training. Asterisks (*) represent significantly lower relative increases with control vs. trained groups. Means ± *SD*s are presented for percentage changes.

Practical Applications

Cross-education training not only promotes strength training improvements in the contralateral homologous muscle but also can enhance training adaptations in heterologous muscles of the upper body with upper-body training. Hence, although the cross-education adaptations may be more effective with youth vs. adults, both groups can benefit. If time is a constraint, then it is important to know that training 1 muscle group can have a relatively global effect, thus reducing the need to specifically train every muscle group (time efficiency). Furthermore, with injury to 1 arm, training the contralateral limb can help to maintain or even increase strength in the untrained muscle group. This study provides helpful information to coaches and other professionals to better understand the cross-education effects for training prescription in situations where it is not possible to perform bilateral training (e.g., immobilization) for youth and adults. In addition, it shows some specific differences between youth and adults that could make an essential difference in professional practice.

Acknowledgments

Funding: No funding is associated with this research project. Conflicts of Interest: The authors declare no conflict of interest. Author contributions: Conceptualization: A. Ben Othman, A. Chaouachi; methodology: A. Ben Othman, A. Chaouachi; data collection: A. Ben Othman, A. Chaouachi; formal analysis, D.G. Behm; writing—original draft preparation: S. Hadjizadeh Anvar, J.C. Aragao-Santos, D.G. Behm; review and editing: S. Hadjizadeh Anvar, J.C. Aragao-Santos, D.G. Behm. All authors have read and agreed to the published version of the manuscript.

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